Title: Student Performance Analysis using Data Analytics

1. Student Details

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2. Problem Statement

Student performance is influenced by multiple factors such as study time, attendance, parental education, internet access, and lifestyle habits. It becomes difficult to identify which factors significantly affect academic results.

The aim of this project is to analyze student performance data to understand key influencing factors and generate useful insights for improving academic outcomes.

3. Domain

Education / Data Analytics

This project belongs to the education domain as it focuses on analyzing student academic performance. It also involves data analytics techniques such as preprocessing, visualization, and insight generation to understand patterns in student data.

4. Dataset Description

Dataset Name: Student Performance Factors Dataset
Source: Kaggle / Provided Dataset
link: "https://www.kaggle.com/datasets/lainguyn123/student-performance-factors"

Features in Dataset:

- Hours Studied
- Attendance
- Parental Involvement
- Access to Resources
- Sleep Hours
- Previous Scores
- Motivation Level
- Internet Access
- Tutoring
- Family Income
- Teacher Quality
- School Type
- Peer Influence
- Physical Activity
- Exam Score (Target Variable)

5. Objectives

- To analyze student performance data
- To identify key factors affecting exam scores
- To study relationships between variables
- To perform data cleaning and preprocessing
- To visualize patterns using graphs
- To generate insights for improving student performance

6. Methodology

1. Data Collection

The dataset is collected from Kaggle or provided source containing student-related attributes.

2. Data Cleaning and Preprocessing

- Handling missing values using mean/mode
- Removing duplicates
- Fixing inconsistent data
- Converting categorical data into proper format

3. Exploratory Data Analysis (EDA)

- Analyzing distribution of exam scores
- Identifying relationships (e.g., study hours vs marks)
- Detecting outliers

4. Data Visualization

- Histogram → score distribution
- Bar chart → category comparison
- Scatter plot → relationship analysis
- Box plot → outlier detection

5. Data Transformation

- Normalization of numeric values
- Encoding categorical variables

6. Insight Generation

- Identify most important factors affecting performance
- Understand trends and patterns

7. Power BI Dashboard

- A dashboard was created using Microsoft Power BI Desktop to visualize the data.
- Dashboard Features:
- KPI Card showing average score
- Bar charts (Study Hours vs Score, Attendance vs Score)
- Pie chart for categorical distribution
- Line chart showing trends
- Slicer for interactive filtering

8. GitHub Repository

- The complete project is uploaded on GitHub, including:
- Python notebook
- Dataset
- Power BI dashboard
- Project documentation

7. Tools & Technologies

- **Python** – Programming
- **Pandas & NumPy** – Data processing
- **Matplotlib** – Visualization
- **Machine Learning** –Linear Regression
- **GitHub**
- **Google Colab** – Execution

8. Expected Outcome

- Understanding of student performance patterns
- Identification of key influencing factors (study hours, attendance, etc.)
- Better insights for improving academic results
- Clear visualization of data trends

9. Conclusion

The project successfully analyzes student performance data using data analytics techniques. Missing values and outliers were handled effectively, and data transformation was applied. The analysis revealed important factors such as study time, attendance, and previous scores that significantly influence exam performance. The results can help in improving student learning strategies and academic outcomes.